Thermodynamics: Principles and Applications

Engineering Thermodynamics — Semester 1

April 18, 2026

Learning Objectives

1. Understand the fundamental laws of thermodynamics.
2. Analyze thermodynamic processes and cycles.
3. Apply the concepts of energy, heat, and work in engineering applications.
4. Solve problems involving ideal gases and phase changes.

1. Introduction to Thermodynamics

Thermodynamics is the branch of physics that deals with the relationships between heat, work, temperature, and energy. It plays a crucial role in various engineering disciplines, including mechanical, chemical, and aerospace engineering. Understanding thermodynamic principles is essential for designing systems such as engines, refrigerators, and power plants.

Definition 1.1. Thermodynamics is the study of energy transformations and the laws governing these transformations in physical systems.

The four fundamental laws of thermodynamics govern the behavior of energy in systems. These laws are essential for analyzing processes and understanding how energy is conserved or transformed.

- **Zeroth Law:** If two systems are in thermal equilibrium with a third system, they are in thermal equilibrium with each other.
- **First Law:** Energy cannot be created or destroyed, only transformed. Mathematically, this is expressed as:
$$\Delta U = Q - W$$
where ΔU is the change in internal energy, Q is the heat added to the system, and W is the work done by the system.
- **Second Law:** The total entropy of an isolated system can never decrease over time. It introduces the concept of irreversibility in natural processes.
- **Third Law:** As the temperature of a system approaches absolute zero, the entropy approaches a minimum value.

Example 1.1: First Law of Thermodynamics

Consider a closed system where 500 J of heat is added, and 200 J of work is done by the system. Calculate the change in internal energy.

$$\Delta U = Q - W = 500\,\mathrm{J} - 200\,\mathrm{J} = 300\,\mathrm{J}.$$

2. Thermodynamic Processes

Thermodynamic processes describe how a system changes from one state to another. These processes can be classified into several types, including isothermal, adiabatic, isobaric, and isochoric processes.

Theorem 2.1 (Isothermal Process)**.** *In an isothermal process, the temperature remains constant ($T =$ constant). For an ideal gas, the relationship between pressure and volume is given by:*

$$PV = nRT,$$

where n is the number of moles and R is the ideal gas constant.

During an isothermal expansion, the system does work on the surroundings, and heat is absorbed to maintain constant temperature. Conversely, in an isothermal compression, work is done on the system, and heat is released.

Example 2.1: Isothermal Expansion of an Ideal Gas

An ideal gas expands isothermally at $T = 300\,\mathrm{K}$ from a volume of $V_1 = 1\,\mathrm{m}^3$ to $V_2 = 2\,\mathrm{m}^3$. Calculate the work done by the gas.

$$W = nRT \ln\left(\frac{V_2}{V_1}\right).$$

Assuming $n = 1\,\mathrm{mol}$ and $R = 8.314\,\mathrm{J/(mol\ K)}$:

$$W = 1 \cdot 8.314 \cdot 300 \ln\left(\frac{2}{1}\right) \approx 1720.8\,\mathrm{J}.$$

Common Mistake

A frequent error is to neglect the heat transfer during isothermal processes. Remember that heat must flow into or out of the system to maintain constant temperature.

3. Applications of Thermodynamics

Thermodynamics has numerous applications in engineering, from designing engines to understanding refrigeration cycles. One common application is the Carnot cycle, which is an idealized thermodynamic cycle that provides the maximum possible efficiency for a heat engine.

Theorem 3.1 (Carnot Efficiency)**.** *The efficiency (η) of a Carnot engine operating between two heat reservoirs at temperatures T_H and T_C is given by:*

$$\eta = 1 - \frac{T_C}{T_H}.$$

This equation highlights the importance of temperature differences in maximizing efficiency. The Carnot cycle consists of two isothermal processes and two adiabatic processes, allowing for the conversion of heat into work.

Example 3.1: Carnot Engine Efficiency

Consider a Carnot engine operating between a hot reservoir at $T_H = 600\,\mathrm{K}$ and a cold reservoir at $T_C = 300\,\mathrm{K}$. Calculate the efficiency.

$$\eta = 1 - \frac{300}{600} = 0.5 \quad \text{or} \quad 50\%.$$

4. Summary

Summary

- Thermodynamics is the study of energy transformations and the laws governing them.
- The four fundamental laws of thermodynamics are essential for analyzing energy systems.
- Different thermodynamic processes (isothermal, adiabatic, etc.) describe how systems change states.
- The Carnot cycle represents the maximum efficiency achievable by a heat engine.

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